Hootan Seifi

AI Engineer · LLM Applications & Agentic AI Systems · Python, FastAPI, LangGraph, RAG

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Live portfolio: hootanseifi.pro | **GitHub:** github.com/hootanseifi | **LinkedIn:** linkedin.com/in/hootan-seifi-42b180124
Open to on-site, remote, hybrid and project-based roles

SUMMARY

AI Engineer specializing in the design, development, evaluation and deployment of LLM-powered systems, including RAG pipelines and AI agents. I have built and deployed LangGraph agent systems, hybrid retrieval, LLM evaluation suites, guardrails and CI/CD infrastructure. My live portfolio, **hootanseifi.pro**, runs a document-grounded RAG question-answering system and a seven-agent LangGraph system that simulates Bitcoin trading on live market prices, under hard AI budget caps. M.Sc. in Electrical Engineering (Artificial Intelligence), with earlier experience in R&D, application review and financial-market analysis.

TECHNICAL SKILLS

LLM & Agents: LangGraph, LangChain, OpenAI Agents SDK, CrewAI, MCP, tool calling, structured outputs, prompt engineering, LLM evaluation, guardrails, QLoRA / PEFT (Hugging Face TRL)

RAG & Retrieval: PostgreSQL + pgvector, hybrid search (vector + BM25, reciprocal rank fusion), embeddings, chunking, reranking, Chroma, Recall@k / MRR

Backend & Deployment: Python, FastAPI, SQLAlchemy, Alembic, REST, Server-Sent Events, Docker Compose, Caddy, Linux, GitHub Actions CI/CD, pytest, Playwright

Machine Learning: PyTorch, scikit-learn, XGBoost, Pandas, SQL; frontend with Next.js and TypeScript

Familiar with: AWS (Bedrock, SageMaker, Lambda), Terraform, Modal, Vercel, Spark, Kafka, Airflow

AI ENGINEERING EXPERIENCE

AI Engineer (Independent Projects) — Self-employed Aug 2026 – Present · Tehran

Design, build, test and deploy LLM and agentic applications end to end. All code is public on GitHub.

hootanseifi.pro — Live AI portfolio: RAG Q&A system, job-fit scoring, AI trading simulation hootanseifi.pro

- Built a bilingual (English/Persian) RAG question-answering system grounded only in my own documents, with source citations. After evaluating weak Persian retrieval, implemented hybrid search (BM25 + pgvector, reciprocal rank fusion). Scored **72/72** on an internal evaluation set that includes refusal and prompt-injection cases.
- Designed a seven-agent LangGraph system that simulates Bitcoin trading using live market prices. Risk rules (max 5x leverage, max 20% margin, 1–5% stop loss) are enforced in Python, not in the prompt: **20/20** evaluated meetings with zero rule violations.
- Built a job-fit scoring system that sets the allowable score range programmatically from must-have skill coverage, preventing overly optimistic model assessments; temperature set to 0 for repeatable outputs.
- Controlled AI inference cost with hard daily and monthly budget caps checked before every model call, and a run-once-per-day design: the first visitor triggers the trading meeting and everyone else watches the recording.
- Deployed 5 containerized services (Next.js, FastAPI, worker, PostgreSQL, Caddy) on a hardened VPS with nightly backups. The GitHub Actions pipeline runs linters, **225 backend, 28 component and 15 end-to-end tests** and a secret scan, and rolls back a failed deploy automatically, verified with a deliberately broken release.

FastAPI · LangGraph · PostgreSQL + pgvector · SQLAlchemy · Next.js / TypeScript · Docker · Caddy · GitHub Actions

Astra — Enterprise AI agent platform github.com/hootanseifi/astra-ai-platform

- Designed a LangGraph agent that dynamically routes each request to RAG, tool calls, both, or a direct answer, with multi-step planning, a recovery node, a PostgreSQL checkpointer and human-in-the-loop approval for sensitive tools.
- Implemented agentic RAG on pgvector (HNSW index, reranking, permission-aware retrieval); reranking raised Recall@1 from **0.67 to 1.00** on the evaluation set.
- Built the security layer (RBAC tool executor with allow-list and schema validation, JWT auth, prompt-injection detection in English and Persian) and operations features (secret redaction, rate limiting, Prometheus metrics, token and cost tracking, an MCP server).
- **407 automated tests, 93% coverage**, and a CI quality gate on Recall@k, MRR, route accuracy and groundedness.

Price Estimate — ML price prediction and deal-finding agents github.com/hootanseifi/price-estimate

- Re-implemented the capstone of Ed Donner's LLM Engineering course as a tested Python package, using an Amazon dataset of about 800,000 products: models from baselines to a deep residual network, a QLoRA fine-tuning pipeline for Llama 3.2 3B (Hugging Face TRL, served on Modal) and an LLM-powered RAG pipeline using Chroma.
- Replaced hand-picked ensemble weights with weights fitted by non-negative least squares, with automatic re-weighting when a model is unavailable; fixed parser, upload and checkpoint bugs from the original notebooks.
- Benchmarked all models on the same test set, reporting MAE with 95% confidence intervals, RMSLE, R^2 and hit rate. Agents monitor deal RSS feeds and send push notifications; served through a FastAPI */predict* endpoint and a Gradio UI, with 60+ offline tests in CI.

AI Medical Notes — SaaS for doctors github.com/hootanseifi/ai-medical-notes-saas

- Turns consultation notes into a structured visit summary, suggested next steps and a draft patient email, streamed to the UI in real time with Server-Sent Events. Next.js, FastAPI, OpenAI, Clerk authentication and subscriptions, deployed on Vercel.

Behengam — Integrated health platform (founder, design phase) Interactive investor demo v2

- Designed to connect 12 stakeholder groups across Iran's healthcare system through a unified patient record. Built the interactive investor demo and the product roadmap: HL7 v2 / FHIR ingestion through Kafka, LOINC / UCUM normalization, Django REST with role-based access, and an AI alert layer whose outputs are shown to doctors as suggestions, never decisions.

EARLIER EXPERIENCE

Digital Marketing & Content Manager — Eghtesad Gostar Behsaman May 2025 – Apr 2026

- Planned and managed digital marketing activities, content production and optimization of communication channels.

Senior Power Plant Specialist — Taban Energy Power Plant (Pasargad Bank) Mar 2025 – May 2025

- Technical responsibilities and oversight of the performance of plant equipment and operational processes.

Fundamental Analyst & Trader, Financial Markets — Novin Eghtesad Technologies Sep 2022 – Feb 2025

- Analyzed financial and crypto markets, evaluated blockchain projects for analytical reports, and developed a Python trading bot on market data.

Technical Specialist, Application Review & Development — Cafe Bazaar Jan 2022 – May 2022

- Technical review of applications, evaluation of their performance quality, and collaboration in the review and publishing process.

R&D Specialist, R&D Department — MAPNA Electric & Control Engineering (MECO) Apr 2020 – Mar 2021

- Contributed to research and development activities and technical review of projects in the company's field.

EDUCATION

M.Sc. Electrical Engineering, Artificial Intelligence — Amirkabir University (Tafresh Branch) 2016 – 2018

B.Sc. Electrical Engineering, Electronics — Amirkabir University (Tafresh Branch) 2011 – 2015

Academic ML projects: fraud detection on Ethereum transactions with imbalanced data (XGBoost); tweet sentiment vs. Nvidia stock price (NLP, time series); traffic-sign classification with CNNs (GTSRB).

CERTIFICATIONS & LANGUAGES

LLM Engineering: Master AI, Large Language Models & Agents — Ed Donner, Udemy

Machine Learning Specialization — Andrew Ng, DeepLearning.AI & Stanford

Financial Trading in Python — DataCamp, 2025

Languages: Persian (native), English (professional working proficiency)